



Imagen 4 for Image Generation

Use text prompts to generate novel images.

In Vertex AI Studio öffnen

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Overview

Imagen 4 for Image Generation brings Google's advanced text-to-image technology to application developers. Imagen boasts a variety of image generation features to help organizations create images that match their specific brand requirements with the same enterprise-grade reliability and governance you're used to with Imagen.

With Imagen for Generation, you can use text prompts to generate novel images. Image generation is delivered by

- Imagen 4 001 model family
 - Imagen 4 001**, the standard model that brings a balanced performance of speed and quality, model version **imagen-4.0-generate-preview-06-06**
 - Imagen 4 Ultra 001**, the slower speed, better quality and more expensive variant **imagen-4.0-ultra-generate-preview-06-06**
 - Imagen 4 Fast 001**, the fast version variant **imagen-4.0-fast-generate-preview-06-06**

For Imagen 4 001, supported languages for text prompts in Imagen Generation include English, Chinese, Hindi, Japanese, Korean, Portuguese, Spanish, German, and French.

Use cases

- Generate images:** Enter text prompts to generate a series of images. For example, "A French cafe with the Golden Gate Bridge in the background."

Documentation

Get started

You can use Imagen for Generation in the Google Cloud console or send in a request to the Vertex AI API.

Generate images using text prompts

Using the console

- Click **Open in Vertex AI Studio**
- In the **Write your prompt** field, enter a description for the images you want to generate. For details about writing effective prompts, see the [Prompt and image attribute guide](#).
- Optional. Change the [parameters](#).
- Optional. You can specify the model version in the **Model options** menu.
- Click **Generate** to generate images

Using the API (curl)

Before using any of the request data, make the following replacements:

- PROJECT_ID:** Your Google Cloud project ID.
- TEXT_PROMPT:** The text prompt guides what images the model generates. This field is required for generation.
- IMAGE_COUNT:** The number of generated images. Accepted integer values: 1-4.

Modell-ID

publishers/google/model-4.0-generate-preview-06-06

Versionsname

google/imagen-4.0-generate-preview-06-06

Tags

Task

Generierung

HTTP method and URL:

```
POST https://us-central1-aiplatform.googleapis.com/v1/projects/PROJECT_ID/locations/us-central1/
```

Request JSON body:

```
{
  "instances": [
    {
      "prompt": "TEXT_PROMPT"
    }
  ],
  "parameters": {
    "sampleCount": IMAGE_COUNT
  }
}
```

Save the request body in a file named request.json and execute the following command:

```
curl -X POST \
  -H "Authorization: Bearer $(gcloud auth print-access-token)" \
  -H "Content-Type: application/json; charset=utf-8" \
  -d @request.json \
  "https://us-central1-aiplatform.googleapis.com/v1/projects/PROJECT_ID/locations/us-central1/
```

You can also set the model version by imagen-4.0-ultra-generate-preview-06-06 or imagen-4.0-fast-generate-preview-06-06 instead of imagen-4.0-generate-preview-06-06 as in the following command:

```
curl -X POST \
  -H "Authorization: Bearer $(gcloud auth print-access-token)" \
  -H "Content-Type: application/json; charset=utf-8" \
  -d @request.json \
  "https://us-central1-aiplatform.googleapis.com/v1/projects/PROJECT_ID/locations/us-central1/
```

```
curl -X POST \
  -H "Authorization: Bearer $(gcloud auth print-access-token)" \
  -H "Content-Type: application/json; charset=utf-8" \
  -d @request.json \
  "https://us-central1-aiplatform.googleapis.com/v1/projects/PROJECT_ID/locations/us-central1/
```

[Learn more about Imagen model versions](#)

Example response

The following sample response is for a request with sampleCount as 2. The response returns two prediction objects, with the generated image bytes base64-encoded.

```
{
  "predictions": [
    {
      "bytesBase64Encoded": "BASE64_IMG_BYTES",
      "mimeType": "image/png"
    },
    {
      "mimeType": "image/png",
      "bytesBase64Encoded": "BASE64_IMG_BYTES"
    }
  ]
}
```

Advanced parameters

When you generate images there are many parameters you can set depending on your use case. The following are some useful parameters you can use. To see the full list, go to the [Imagen for image generation API reference](#)

- **Prompt enhancement** (enhancePrompt, available for Imagen 4 model family): Let Vertex AI backend to enhance the input prompt with an LLM-based prompt rewriting tool. Enabled by default. Set to false to disable prompt enhancement.

Versions

Resource ID	Release date	Release stage	Description
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imagen-4.0-generate-preview-06-06	2025-06-06	Public Preview	Imagen 4: Support Generation with 1:1, 9:16, 16:9, 3:4, 4:3 aspect ratios and watermarking (SynthID).
imagen-4.0-ultra-generate-preview-06-06	2025-06-06	Public Preview	Imagen 4 Ultra: Support Generation with 1:1, 9:16, 16:9, 3:4, 4:3 aspect ratios and watermarking (SynthID).
imagen-4.0-fast-generate-preview-06-06	2025-06-06	Public Preview	Imagen 4 Fast: Support Generation with 1:1, 9:16, 16:9, 3:4, 4:3 aspect ratios and watermarking (SynthID).
imagen-4.0-generate-preview-05-20	2025-05-20	Public Preview	Imagen 4: Support Generation with 1:1, 9:16, 16:9, 3:4, 4:3 aspect ratios and watermarking (SynthID).
imagen-4.0-ultra-generate-preview-05-20	2025-05-20	Public Preview	Imagen 4 Ultra: Support Generation with 1:1, 9:16, 16:9, 3:4, 4:3 aspect ratios and watermarking (SynthID).
imagen-3.0-generate-002	2025-01-23	GA	Imagen 3: Support Generation with 1:1, 9:16, 16:9, 3:4, 4:3 aspect ratios and watermarking (SynthID).
imagen-3.0-generate-001	2024-07-31	GA	Imagen 3: Support Generation with 1:1, 9:16, 16:9, 3:4, 4:3 aspect ratios and watermarking (SynthID).
imagen-3.0-fast-generate-001	2024-07-31	GA	Imagen 3 fast: Support Generation of lower latency version with 1:1, 9:16, 16:9, 3:4, 4:3 aspect ratios and watermarking (SynthID).
imagegeneration@006	2024-04-09	GA	Imagen 2: Supports Generation with 1:1, 9:16, 16:9, 3:4, 4:3 aspect ratios and watermarking (SynthID). Supports editing with four editing modes
imagegeneration@005	2023-12-11	GA	Imagen 2: Supports generation with 1:1 and 9:16 aspect ratios
imagegeneration@002	2023-07-17	GA	Imagen 1: GA version that supports both generation and editing features

Links

- [Imagen on Vertex AI overview](#)
- [Generate images using text prompts](#)
- [Edit images using text prompts](#)
- [Usage guidelines](#)
- [Prompt and image attribute guide](#)

Pricing

- [Pricing for the Imagen models](#)